

Softlend Fintech Backend Assignment

API Testing and Database Validation Report

1. Customer Database Verification

Objective

To verify successful customer creation and credit score storage within the database.

Endpoint Tested

Customer Creation and Credit Score Update Module

Validation Performed

- Customer record created successfully.
- Unique customer identifier generated.
- Customer information stored correctly.
- CIBIL score updated successfully.
- Score fetch timestamp recorded.

Result: Passed

Outcome

The customer information was successfully persisted in the MySQL database and is available for subsequent credit assessment, offer generation, and analytics operations.

2. Create Credit Gap API

Endpoint

POST /api/v1/customers/{customerId}/credit-gaps

Objective

To identify and record credit improvement opportunities for a customer.

Validation Performed

- Credit gap record created successfully.
- Credit factor recorded.

- Current and ideal values stored.
- Impact level assigned.
- Estimated score improvement calculated.
- Corrective action recommendation generated.

Result: Passed

Outcome

The system successfully identified credit weaknesses and generated actionable recommendations to improve the customer's credit profile.

3. Get Credit Profile API

Endpoint

GET /api/v1/customers/{customerId}/credit-profile

Objective

To retrieve a complete customer credit profile, including associated improvement opportunities and offers.

Validation Performed

- Customer details retrieved successfully.
- Credit gaps retrieved successfully.
- Available offers retrieved successfully.
- Potential credit score calculated.
- Improvement summary generated.

Result: Passed

Outcome

The API successfully consolidated customer information, credit gaps, offer data, and projected score improvements into a comprehensive profile response.

4. Improvement Summary API

Endpoint

GET /api/v1/customers/{customerId}/improvement-summary

Objective

To provide a summarized analysis of customer credit improvement opportunities.

Validation Performed

- Current credit score retrieved.
- Total estimated score gain calculated.
- Impact categories identified and counted.
- Gap factors summarized.

Result: Passed

Outcome

The API successfully generated a concise summary of credit improvement opportunities and their expected impact on the customer's overall creditworthiness.

5. Customer Statistics API

Endpoint

GET /api/v1/customers/stats

Objective

To generate analytical insights from customer data.

Validation Performed

- Total customer count calculated.
- Customers with valid credit scores identified.
- Average credit score computed successfully.

Result: Passed

Outcome

The API successfully generated customer-level analytics suitable for business reporting and decision-making.

6. Create Offer API

Endpoint

POST /api/v1/offers/customers/{customerId}/offers

Objective

To create and assign a loan offer to an eligible customer.

Validation Performed

- Offer created successfully.
- Loan amount stored.
- Interest rate recorded.
- Loan tenure stored.
- Minimum score requirement validated.
- Initial offer status assigned.

Result: Passed

Outcome

The system successfully generated and stored a loan offer associated with the selected customer profile.

7. Get Customer Offers API

Endpoint

GET /api/v1/offers/customers/{customerId}/offers

Objective

To retrieve all loan offers associated with a customer.

Validation Performed

- Customer offers retrieved successfully.
- Offer details returned accurately.
- Offer status displayed correctly.

Result: Passed

Outcome

The API successfully returned all loan offers available for the specified customer.

8. EMI Calculation API

Endpoint

GET /api/v1/offers/{offerId}/emi

Objective

To calculate the Equated Monthly Installment (EMI) for a loan offer.

Validation Performed

- Loan amount retrieved successfully.
- Interest rate retrieved successfully.
- Loan tenure retrieved successfully.
- EMI calculated accurately.

Result: Passed

Outcome

The system successfully calculated the monthly repayment amount based on the loan amount, interest rate, and repayment tenure.

9. Update Offer Status API

Endpoint

PATCH /api/v1/offers/{offerId}/status

Objective

To manage and update the lifecycle status of a loan offer.

Validation Performed

- Offer status updated successfully.
- Database record modified correctly.
- Updated timestamp recorded.

Result: Passed

Outcome

The system successfully handled offer lifecycle transitions, including Pending, Active, and Disbursed states.

10. Offer Statistics API

Endpoint

GET /api/v1/offers/stats

Objective

To provide analytical insights related to loan offers.

Validation Performed

- Total offers counted successfully.
- Active offers counted successfully.
- Pending offers counted successfully.
- Disbursed offers counted successfully.

Result: Passed

Outcome

The API successfully generated statistical information regarding offer distribution and status management.

Technology Stack

- Node.js
- Express.js

- MySQL
- Sequelize ORM
- RESTful API Architecture
- Postman API Testing Tool

Database Tables Utilized

1. Customers
2. Credit_Gaps
3. Offers

Assignment Completion Status

- ✓ Customer Management Module Completed
- ✓ Credit Gap Analysis Module Completed
- ✓ Credit Profile Management Module Completed
- ✓ Improvement Summary Module Completed
- ✓ Customer Analytics Module Completed
- ✓ Offer Management Module Completed
- ✓ EMI Calculation Module Completed
- ✓ Offer Analytics Module Completed
- ✓ Database Integration Completed
- ✓ API Validation and Testing Completed